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Antibiotic resistance increases with local temperature

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Bacteria that cause infections in humans can develop or acquire resistance to antibiotics commonly used against them[1,2]. Antimicrobial resistance (in bacteria and other microbes) causes significant morbidity worldwide, and some estimates indicate the attributable mortality could reach up to 10 million by 2050[2–4]. Antibiotic resistance in bacteria is believed to develop largely under the selective pressure of antibiotic use; however, other factors may contribute to population level increases in antibiotic resistance[1,2]. We explored the role of climate (temperature) and additional factors on the distribution of antibiotic resistance across the United States, and here we show that increasing local temperature as well as population density are associated with increasing antibiotic resistance (percent resistant) in common pathogens. We found that an increase in temperature of 10 °C across regions was associated with an increases in antibiotic resistance of 4.2%, 2.2%, and 2.7% for the common pathogens *Escherichia coli*, *Klebsiella pneumoniae* and *Staphylococcus aureus*. The associations between temperature and antibiotic resistance in this ecological study are consistent across most classes of antibiotics and pathogens and may be strengthening over time. These findings suggest that current forecasts of the burden of antibiotic resistance could be significant underestimates in the face of a growing population and climate change[4].

Since the discovery of penicillin by Alexander Fleming in 1928, antibiotics have had an immeasurable impact on the reduction of bacterial infections and associated morbidity and mortality[1,2]. However, their global use (and overuse) has supported the spread of bacterial resistance to these agents[1,2], with the possibility of entering a 'post-antibiotic era' in which antibiotics are no longer a treatment option[2–4]. Despite the potentially dire implications for human health posed by antibiotic resistance over the next century, we have a limited understanding of the factors, including the potential impact of climate change, that drive the transmission of antibiotic resistance among populations[1,5].

Antibiotic resistance can spread by the horizontal transmission of specific genomic resistance mechanisms and the selection/propagation of resistant strains[1,6], which is known to occur (often asymptomatically) between humans, animals and the environment[1] (we collectively refer to this movement as 'transmission'). Although antibiotic use has been assumed to be the main force behind the rise of antibiotic resistance in populations, other factors may act to facilitate the transmission of antibiotic resistance and drive regional spread[1,7]. Temperature has long been known to affect bacterial growth in vitro as well as modulate the transfer of genomic material, which includes genes that encode (or confer) antibiotic resistance[8,9], but empirical evidence of the impact of climate on population level resistance is lacking. Here we investigated whether local climate, in the form of local temperature, could explain differences in antibiotic resistance across geographical regions in the United States at a high spatial resolution. In addition, we sought to evaluate whether local characteristics, such as population density, antibiotic prescription rates, specific patient populations and laboratory standards could also account for observed differences in antibiotic resistance patterns.

We developed a large database of antibiotic resistance patterns across different geographical regions in the United States using antibiotic resistance indices (by antibiotic and bacterial pathogen) generated from information sourced from hospitals, laboratories and surveillance units[10]. Resistance indices were linked with facility characteristics, such as inpatient/outpatient type, as well as regional prescribing rates[11] and population density[12]. We studied three common community- and hospital-acquired bacterial pathogens, including two Gram-negative species (*E. coli* and *K. pneumoniae*) and one Gram-positive species (*S. aureus*)[13,14]. The final data set represented over 1.6 million clinically relevant bacterial pathogens from 602 unique indices that spanned 223 facilities across 41 states for the years 2013–2015.

We explored the relationships between the regional patterns of antibiotic resistance (here defined as percent of pathogens not susceptible to a particular antibiotic) and both latitude and local historical climate variables (mean and minimum temperature, averaged over 1980–2010) (Supplementary Fig. 1)[15]. Historical climate variables were chosen to match best with the periods of time over which antibiotic resistance patterns have developed. Average minimum temperature showed higher correlations than average temperature and latitude across the three pathogens and antibiotics (Supplementary Fig. 1), with correlations up to 0.7. Minimum temperature is a commonly used climate variable when describing the survival of species within environments[16,17] and may be of particular importance to identify: (1) regions that can support the persistence of environmental and/or colonizing bacteria and (2) regions with an enhanced growth and resistance transmission potential. Given these factors, and the relative strength of its relationship with antibiotic resistance, we focused our analysis on the association between minimum temperature and antibiotic resistance.

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Using both unadjusted and multivariable-adjusted weighted linear models, we evaluated multiple aspects of the association between minimum temperature and antibiotic resistance. We found that increasing the minimum temperature was associated with increasing antibiotic resistance, and this was consistent across most classes of antibiotics and pathogens (Figs. 1–2 and Supplementary Figs. 2–4). In the unadjusted analysis (Table 1), an increase of 10 °C across regions was associated with an increase in antibiotic resistance of 5.1% ($P<0.0001$), 3.4% ($P<0.0001$) and 3.1% ($P=0.002$) for *E. coli*, *K. pneumoniae* and *S. aureus*, respectively. Notably, changes in minimum temperature lead to larger increases in antibiotic resistance in subsequent years of observations, when analysing all antibiotics and pathogens, both for oral and intravenous (i.v.) formulations (Fig. 2). The measure of association between minimum temperature and antibiotic resistance may increase with the earlier time of introduction in the United States (Supplementary Fig. 5). The geographical pattern of antibiotic resistance in the common pathogen *E. coli* is shown in Fig. 1, and in Supplementary Figs. 3 and 4 for *K. pneumoniae* and *S. aureus*, respectively.

Multivariable analysis of antibiotic resistance versus minimum temperature was performed for each pathogen across all tested antibiotics (Table 1). After adjusting for acquisition source, prescription rate (available for a subset of tested antibiotics), population density and laboratory standard, we found that a 10 °C increase in minimum temperature across the regions was associated with an increase in antibiotic resistance of 4.2% ($P<0.0001$), 2.2% ($P<0.0001$) and 2.7% ($P=0.21$) for *E. coli*, *K. pneumoniae* and *S. aureus*, respectively. When we removed penicillin (an uncommon and sporadically tested antibiotic) susceptibility results for *S. aureus*, we found that the minimum temperature was associated with an increase in antibiotic resistance of 5.1% ($P<0.0001$) (Table 1). We also looked at *S. aureus* resistance to specific antibiotics, which included cloxacillin, an older antibiotic that is used to differentiate methicillin-susceptible from methicillin-resistant *S. aureus*. In multivariable-adjusted analysis, we found increases in antibiotic resistance to cloxacillin, fluoroquinolones and macrolides of 5.8% ($P<0.0001$), 3.7% ($P=0.096$) and 6.0% ($P<0.001$), respectively, for a 10 °C increase in minimum temperature across regions (Supplementary Table 1). These multivariable results generally support the univariate associations seen.

Geographical variability in antibiotic prescribing has been documented in outpatient facilities between broad regions within

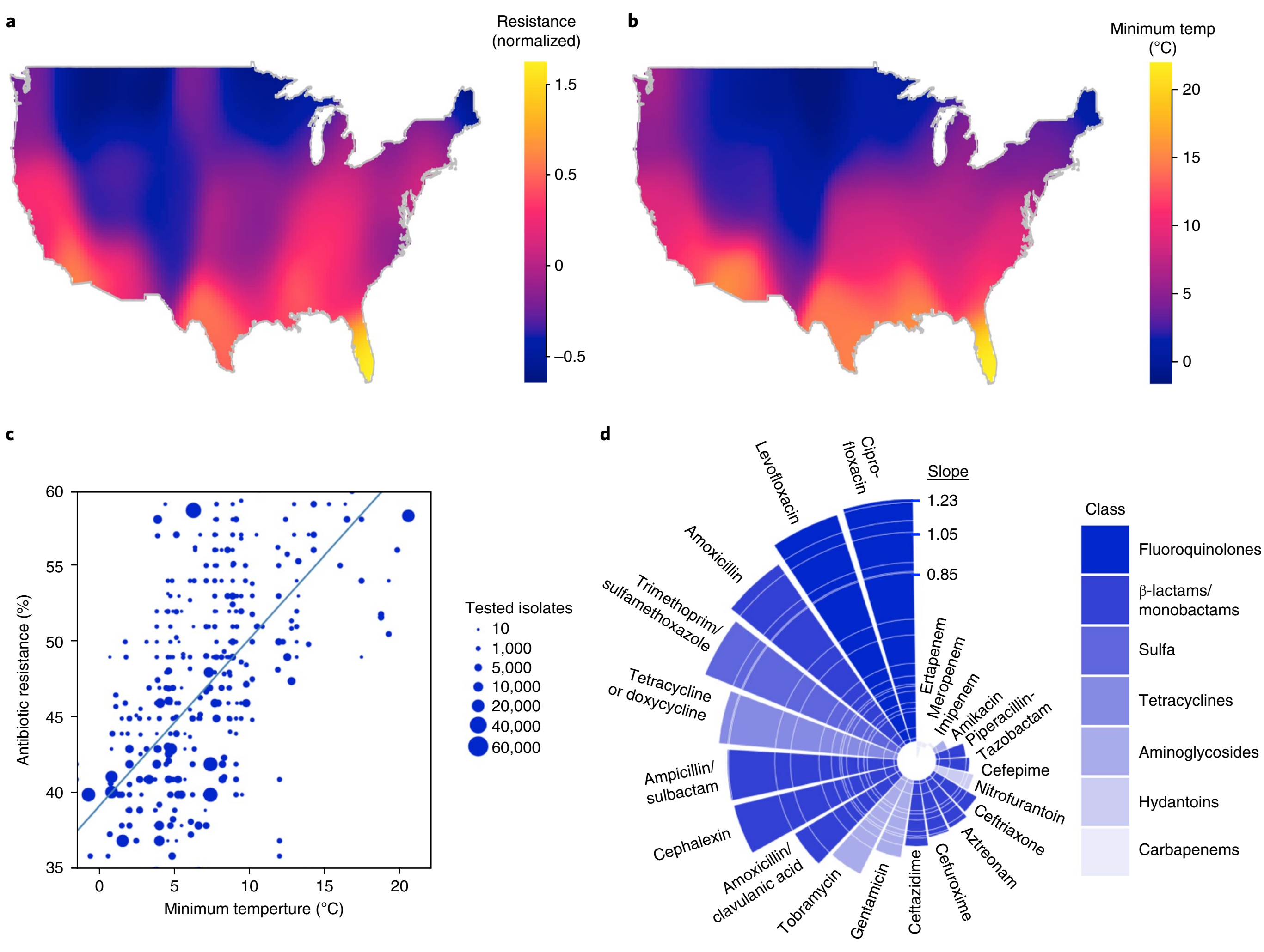

Fig. 1 | Antibiotic resistance increases with increasing temperature. a, A heat map of the mean normalized antibiotic resistance for *E. coli* for all antibiotics across the United States. **b**, A heat map of the 30-year average minimum temperature (°C) across the United States. **c**, A scatter plot of antibiotic resistance vs minimum temperature (°C) for *E. coli* and amoxicillin. The unadjusted weighted linear trend line is shown in blue. **d**, Slope of unadjusted relationship (% resistance per °C) between minimum temperature and antibiotic resistance by antibiotic for *E. coli*. The antibiotic class is coded by colour shading.

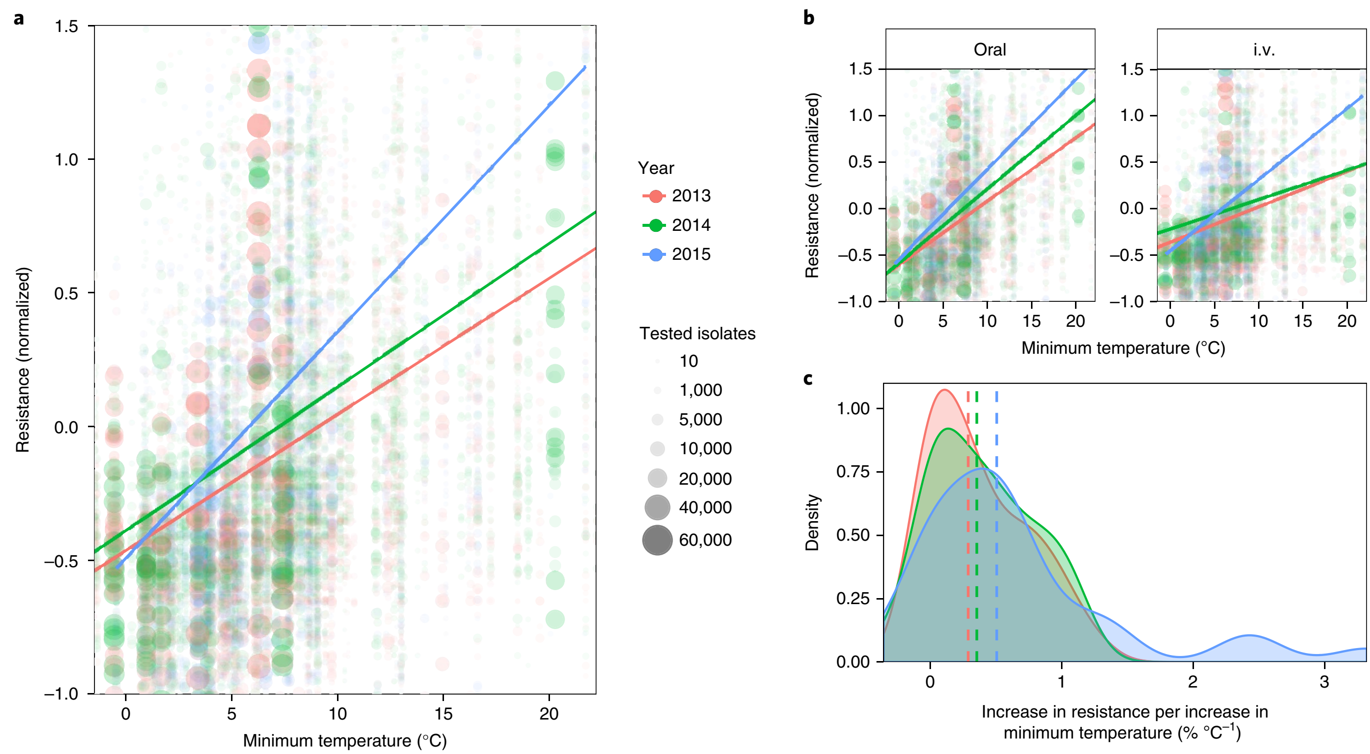

Fig. 2 | Change in the relationship between minimum temperature and antibiotic resistance over time. a, Mean normalized antibiotic resistance versus minimum temperature (°C) for all pathogens and antibiotics, stratified by year (2013–2015). Unadjusted weighted linear relationships for the years 2013–2015 are shown. **b**, Mean normalized antibiotic resistance versus minimum temperature (°C) for all pathogens and antibiotics, stratified by year and route (oral versus i.v.). **c**, Density distributions of association measures (slopes) between antibiotic resistance and minimum temperature, stratified by time and with median densities (by year) marked by vertical dashed lines.

the United States[18,19], with an increased incidence of antibiotic prescriptions observed across a number of southern states[18]. Given these patterns, we examined the effects of spatial distributions of antibiotic prescribing on resistance for four common antibiotic classes, penicillins, cephalosporins, fluoroquinolones and macrolides. We found that increased prescribing was associated with increasing antibiotic resistance for the antibiotics evaluated across all three pathogens (Table 1 and Supplementary Fig. 6). Moreover, the relationship between minimum temperature and antibiotic use did not change by the amount (tertile) of antibiotic prescribing (Supplementary Fig. 7).

In the multivariable-adjusted analyses (Table 1), we found that an increase in population density by 25,900 persons per km^2 (10,000 persons per mile^2), for example, from Bismarck, North Dakota, to Boston, Massachusetts, was associated with a 6% increase in antibiotic resistance for *K. pneumoniae* ($P<0.0001$) and 3% for *E. coli* ($P=0.086$). This may reflect increased transmission rates within dense populations[20]. Sources that contain outpatient antibiotic resistance data were associated with a lower resistance for all the pathogens, and this association was significant in adjusted models for *S. aureus* (−4.24%, $P=0.0003$), but not for *E. coli* (−2.98%, $P=0.06$) or *K. pneumoniae* (−2.88%, $P=0.17$) (Table 1). Reported laboratory standard only demonstrated a significant association with antibiotic resistance in the adjusted analyses of *S. aureus* isolates. We found no change in the relationship between minimum temperature and antibiotic resistance across the levels of the other evaluated covariates (Supplementary Fig. 7). Furthermore, we found no strong correlations between minimum temperature and other covariates (Supplementary Table 2).

Although potential mechanisms that underlie the observed associations between resistance and temperature require further elucidation, several hypotheses are proposed. First, temperature may facilitate horizontal gene transfer, which occurs through the exchange of resistance genes (for example, plasmid-borne extended-spectrum β-lactamases (ESBLs)) or the uptake of free genetic material[8,9,21]. European countries in the southern latitudes have a higher incidence of infections due to the ESBL-producing Enterobacteriaceae[22], which has generally been attributed to antibiotic use and selective pressure, but which may be facilitated by climate factors, including temperature. Many recent emergences of highly mobile genetic elements of resistance have originated from central latitudes[9,23,24]. Increased gene transfer (which occurs either at sites of colonization on a host[25] or in the environment[9]) might then be expected to facilitate population transmission. Second, temperature is one of the most potent modifiers of bacterial growth rates[25,26], and may drive increased carriage[27] and transmission of resistant strains between humans and animals. For both Gram-positive and Gram-negative human pathogens, seasonal patterns of infection have been identified and carriage may play a role[28,29]. In a similar fashion, temperature may support the environmental growth of resistant strains and lead to an enhanced transmission from food, agriculture and environmental sources[30]. Increased population-level transmission may thus facilitate population-level selection of resistant strains[31]. Interestingly, we also found that fluoroquinolones and β-lactams generally demonstrated the strongest associations between temperature and resistance (Fig. 1 and Supplementary Figs. 3 and 4). This supports mechanism-specific impacts of temperature on resistance. Last, the potential temperature associations may be rooted in more complex factors (for example, behavioural and social) that occur across humans, animals, agriculture and environment, as embodied by the One Health perspective. Although

Table 1 | Weighted unadjusted and adjusted (multivariable) regression analysis of evaluated predictors of antibiotic resistance

Bacteria	Variable	Unadjusted	Unadjusted[a]	Adjusted[a]
E. coli	Minimum temperature (°C)	0.512 (<0.0001)	0.598 (<0.0001)	0.419 (<0.0001)
	Prescription rate (per 1,000 population)	-	0.120 (<0.0001)	0.121 (<0.0001)
	Outpatient	−3.280 (0.14)	−4.590 (0.09)	−2.980 (0.06)
	Lab standard	−4.240 (0.08)	−4.900 (0.06)	−3.240 (0.16)
	Population density (persons per 1.26 km^2 (per $mile^2$))	0.00027 0.006	0.00035 (<0.0001)	0.00027 (0.086)
K. pneumoniae	Minimum temperature (°C)	0.341 (<0.0001)	0.397 (<0.0001)	0.219 (<0.0001)
	Prescription rate (per 1,000 population)	-	0.030 (<0.0001)	0.034 (<0.0001)
	Outpatient	−2.640 (0.17)	−3.006 (0.24)	−2.880 (0.17)
	Lab standard	−1.760 (0.17)	−2.220 (0.22)	−0.371 (0.75)
	Population density (persons per 1.26 km^2 (per $mile^2$))	0.00051 (<0.0001)	0.00063 (<0.0001)	0.0006 (<0.0001)
S. aureus	Minimum temperature (°C)	0.305 (0.002)	0.571 (0.004)	0.273 (0.21)
	Prescription rate (per 1,000 population)	-	0.133 (<0.0001)	0.134 (<0.0001)
	Outpatient	−5.800 (<0.0001)	−5.570 (<0.0001)	−4.243 (0.0003)
	Lab standard	−1.780 (0.08)	−3.450 (0.08)	−3.922 (0.03)
	Population density (persons per 1.26 km^2 (per $mile^2$))	0.00014 (0.15)	0.00015 (0.07)	0.00005 (0.69)
S. aureus[b]	Minimum temperature (°C)	0.405 (<0.0001)	0.762 (<0.0001)	0.514 (<0.0001)
	Prescription rate (per 1,000 population)	-	0.068 (0.0006)	0.066 (<0.0001)
	Outpatient	−6.338 (0.0005)	−6.520 (<0.0001)	−4.570 (0.01)
	Lab standard	−2.305 (<0.0001)	−4.910 (0.008)	−3.840 (0.01)
	Population density (persons per 1.26 km^2 (per $mile^2$))	0.0002 (0.035)	0.0002 (0.009)	0.00004 (0.27)

Unadjusted and adjusted (multivariable) regression parameters for historical (30-year) mean minimum temperature, antibiotic class-specific prescription rate (annual prescriptions per 1,000 population), acquisition source (containing outpatient isolates versus inpatient only), laboratory standard (CLSI versus other/not reported) and population density are given, with *P* values in parentheses. [a]Restricted to the subset of antibiotic susceptibilities for dates and classes of antibiotics that correspond to the available prescribing-rate data. [b]Analysis with penicillin susceptibility removed.

a major limitation of our approach is that we compare climate and predictor data with antibiotic resistance at regional levels, and thus cannot infer causality, our approach is suited to identifying ecological associations that may be particularly relevant for antibiotic resistance (for example, antibiotic prescriptions across population versus individual) and is a commonly used design when antibiotic resistance is studied over large populations and geographies[22].

The World Health Organization has identified climate change as a major driver of emerging infectious diseases globally[5], although vector-borne and enteric infections, such as cholera, have typically been considered as the most likely to be impacted. Our findings suggest that the spread of antibiotic resistance may be modified and potentially accelerated by regional temperature and future climate change. Based upon our findings, a 10 °C increase in temperature, an extreme but conceivable scenario for some parts of the

United States by the end of this century[32], could yield additional increases in resistance on the order of 10% for certain antibiotics (Fig. 1 and Supplementary Table 1). If the relationship between minimum temperature and antibiotic resistance is, indeed, present and increasing over time, this could support a more rapid progression towards a 'post-antibiotic era'. Further research is needed to confirm these relationships and elucidate the role of climate in antibiotic resistance spread. Our findings suggest that, in the presence of climate change and population growth, already dire predictions of the impact of antibiotic resistance on global health may be significant underestimates[4].

Methods

Methods, including statements of data availability and any associated accession codes and references, are available at https://doi.org/10.1038/s41558-018-0161-6.

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Author contributions

D.R.M., S.F.M. and M.S. contributed to the data analysis. All the authors (D.R.M., S.F.M., D.F., M.S. and J.S.B.) contributed to development of the manuscript, discussion and preparation of final versions. All the authors approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Methods

Patterns of antibiotic resistance. To identify regional patterns of antibiotic resistance across the United States, we utilized a continuously updated database of spatially localized annual indices of inpatient and outpatient antibiotic resistance for common clinically relevant bacterial pathogens from human clinical isolates for the years 2013–2015[10]. Data entry into this database was performed by analysts blind to the research questions evaluated in this study. Each antibiotic resistance index corresponded to a particular health and/or surveillance network, hospital or laboratory facility. Relevant variables captured in this database include antibiotic susceptibility, bacterial species, year, location, acquisition sources (inpatient only or containing outpatient isolates) and laboratory standard (Clinical and Laboratory Standards Institute (CLSI) or other/not reported). Percent antibiotic resistance was defined as the percentage of tested bacterial isolates that were non-susceptible to a given antibiotic. Minimum inhibitory concentration values were not available and/or evaluated in this data set. Antibiotics evaluated for each pathogen are shown in Supplementary Table 3, including classification as i.v. formulation or available in a regularly used oral formulation. For specific drugs where the testing of i.v. versions approximate the susceptibility for oral equivalents, we substituted the oral drug names (which are more commonly used). These included amoxicillin and cephalexin in lieu of ampicillin and cefazolin. We restricted our analysis to the continental United States and focused on three important pathogens in the community and nosocomial infections, *E. coli, K. pneumoniae* and *S. aureus*[13,14]. The distributions of pathogen-specific data points are shown in Supplementary Fig. 8. After restricting to these three bacterial species, the final data set represented over 1.6 million bacterial isolates that correspond to up to 22.8 million susceptibility test results from 602 unique indices, and span 223 facilities across 41 states throughout the continental United States. The number of indices that contributed data for the pathogens *E. coli*, *K. pneumoniae*, and *S. aureus* were 595, 551 and 307 respectively, which represent approximately 1.1 million, 0.2 million and 0.3 million bacterial pathogens tested, respectively. Of this data set, the mean population density was 6,377 persons per km^2 (2,462 persons per mile^2), 77.4% of indices contained outpatient data and 70.8% of indices reported using CLSI standards (versus other/not reported). For the analysis of the impact of prescription rates, we used a subset of the data based on the availability of prescribing rates for relevant drug classes (fluoroquinolones, cephalosporins, penicillins and macrolides), and this subset contained 446 indices across 40 states, with a total of 1.3 million bacterial pathogens, with a mean prescribing rate of 139.8 prescriptions per 1,000 persons, mean population density of 6,690 persons per km^2 (2,583 persons per mile^2), 78% of indices contained outpatient data and 70% of indices reported using CLSI standards (versus other/not reported). A validation assessment of these data sets was performed, with a review of the antibiotic resistance and descriptor data values randomly selected across indices from a 10% sample of facilities included in the database, and yielded a concordance of 97% with the original data sources.

Predictors and confounders. We selected relevant predictors and confounders that help predict antibiotic resistance based on previous literature and plausible biological mechanisms[1,13,19,26,29,33]. We linked each location by the nearest zip code with climate data, including 30-year normals (1980–2010) of minimum temperature (30-year average of daily minimum temperature) and mean temperature (30-year average of mean daily temperature) from the US National Climatic Data Center[15]. To account for spatial variability in antibiotic use, outpatient antibiotic prescriptions (annual prescriptions per 1,000 persons) by class (fluoroquinolones, cephalosporins, penicillins and macrolides) for the years 2013–2014 were linked to index location by the antibiotic tested at the state level, using data derived from Xponent database from QuintilesIMS via the Centers for Disease Control and Prevention (CDC) Antibiotic Resistance Patient Safety Atlas[11]. We assume that current distributions of antibiotic prescription rates are reflective of recent historical distributions as complete prescription data over a potentially relevant time course are not reasonably identifiable and/or attainable. Moreover, we assume that the prescribing measures and linear relationship we use adequately approximates the underlying relationships, and that we have captured the most relevant predictors/confounders. Finally, we linked the data with population density throughout the United States by zip code based on 2010 census data[12].

Data analysis. For all the figures and analyses in this ecological study, we used location/facility-specific data points, prepared by weighted averages (by number of tested isolates) of antibiotic resistance indices by organism, drug, acquisition sources (inpatient or outpatient containing) and year for the particular location. Where applicable, data markers were scaled proportionally to the total number of isolates tested for a given organism at a particular location/facility and this is denoted in the legends. To ease visualization of the trend of the antibiotic resistance across multiple different antibiotics, we centred the data about the mean and normalized by the standard deviation (Fig. 2 and Supplementary Figs. 2, 6 and 7). Trend bars represent weighted linear-fitted models. The association of minimum temperature and antibiotic resistance was plotted within levels of the covariates (prescription rate—tertile, acquisition location, laboratory standard, and population density—tertile) to assess qualitatively for potential confounding and effect modifications (Supplementary Fig. 7). Log scales were utilized for the population density given the large spread of values (Supplementary Fig. 6). All the plots were generated using the 'ggplot2' package in R (Version 3.3.2).

We generated heat maps of the normalized antibiotic resistance (Fig. 1 and Supplementary Figs. 3 and 4) for each pathogen, based on average normalized resistance within zip codes that contained antibiotic resistance data. We utilized the reference 30-year minimum temperature normals for these same zip codes to generate the minimum temperature map (Fig. 1 and Supplementary Figs. 3 and 4). Gaussian kernel smoothing was performed with a least-squares cross-validated bandwidth selection. Map projections were obtained using the R packages 'maptools' and 'maps', and visualizations were performed using the package 'spatstat'.

We used weighted linear regression of location specific antibiotic resistance with clustering by state and latitude quintile to account for correlated errors, and determined the unadjusted and adjusted model parameters for relevant predictors (Table 1). Weighting was based on the total number of isolates tested for a given pathogen. All the analyses were performed using R (Version 3.3.2).

Sensitivity analyses. To assess the potential impact of possible 'double counting' that could occur in regions in which a large surveillance body may overlap with a hospital or laboratory, we evaluated the association of minimum temperature and antibiotic resistance using multivariable weighted regression, but restricted to the subset of health system, hospital and laboratory data only. There were no major changes in overall interpretation of the data, and the results are shown in Supplementary Table 4.

To evaluate whether time trends in the association of minimum temperature on antibiotic resistance could be due to non-random sampling, we applied the same multivariable regression analysis as described in the main paper, but restricted to institutions that provided data for more than one year. There were no major changes in the overall interpretation of the data, and the multivariable model results are shown in Supplementary Table 4.

To assess whether different isolate sources could impact the association between antibiotic resistance and temperature, we ran additional multivariable linear weighted regression models with the same parameters as the full model in Table 1 (minimum temperature, prescription rate, outpatient containing, laboratory standard and population density) but also included binary predictors of isolate sources contained within the indices, namely: urine, blood, respiratory, other sterile (for example, cerebrospinal fluid, pleural fluid and peritoneal fluid) and other non-sterile (for example, wound swabs). To accommodate these additional predictors, this model was run in the data set prior to collapsing across the variables of interest (see above). There were no major changes in the overall interpretation of the data, and the results are shown in Supplementary Table 4.

To evaluate whether socioeconomic status could have an impact on the association between antibiotic resistance and temperature, we ran additional multivariable weighted linear regression models with the same parameters as the full model, but with either the replacement of population density with median income or the addition of median income (Supplementary Table 4). We found no major changes to the overall model findings, although median income may be a relevant factor for *E. coli* resistance. Median income data were extracted from the US Census Bureau American Census Survey, 5-year estimate from 2015, and linked by zip code. To evaluate the possible impact of imputation of cephalexin susceptibility based on cloxacillin results, we ran additional multivariable weighted linear regression models for *S. aureus* with cephalexin susceptibilities removed. We found no major changes to the overall interpretation of the results (Supplementary Table 4).

Limitations. As previously noted, due to the nature of this descriptive ecological study, and potential for ecologic bias, we cannot infer causality. We have selected confounders based on their potential relevance in antibiotic resistance transmission; however, residual and/or unmeasured confounding is possible. To select important factors that govern the population-level distribution of antibiotic resistance is challenging, as they are generally poorly understood. We also only included prescribing rates for antibiotic classes matching the antibiotic susceptibility, and did not consider the selection effects of other classes given the complexity and uncertainty of their potential impact. Finally, our data set may be subject to selection bias, although it is unlikely that the minimum temperature would be associated with inclusion into the data set.

Data availability. Antibiotic resistance data are derived from available online sources and are queryable at www.resistanceopen.com[10]. Climate data are available through the US National Climatic Data Center[15]. Antibiotic prescription rates are available from Xponent database from QuintilesIMS (via the CDC Antibiotic Resistance Patient Safety Atlas[11]). Population density values are available through the 2010 census data[12].

References

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